

Kafka POC

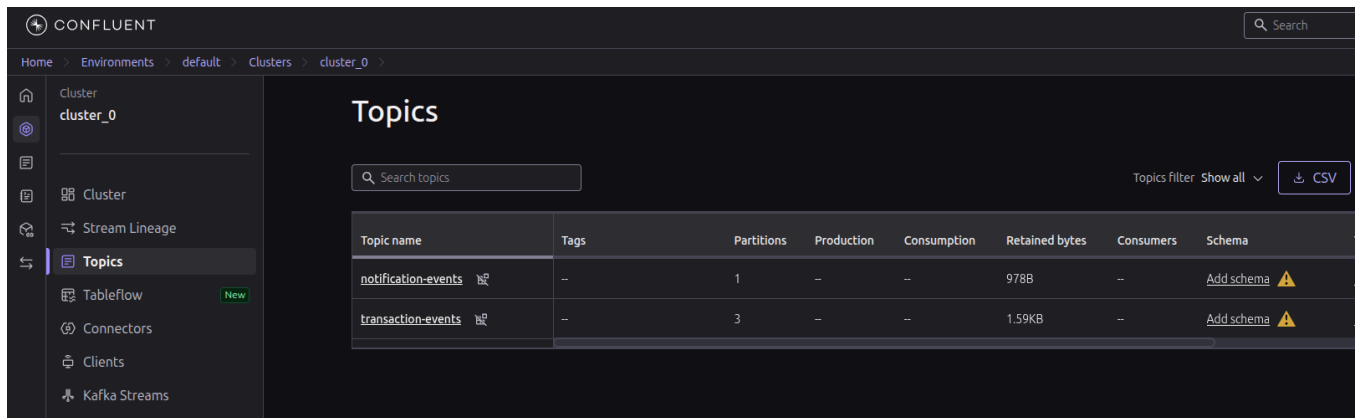
Spring Boot + Spring Kafka | Confluent Cloud

1. POC Overview

- REST API publishing transaction events (approved / initiated / failed) to transaction-events topic.
- Audit consumer on transaction-events -> logs all events; approved events trigger SMS notification.
- Notification consumer on notification-events -> simulates SMS delivery.
- Topics auto-created on Confluent Cloud (transaction-events: 3 partitions, notification-events: 1).
- Partition key = folioNumber for ordered processing per folio.

2. Implementation

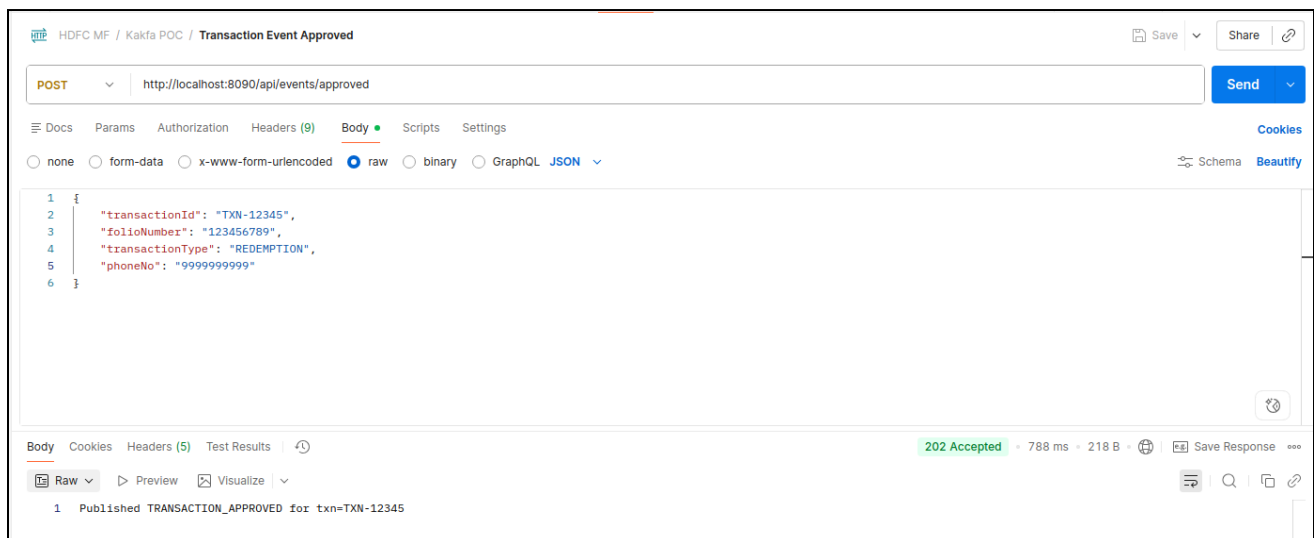
2.1. Confluent Topics



Topic name	Tags	Partitions	Production	Consumption	Retained bytes	Consumers	Schema
notification-events	--	1	--	--	978B	--	Add schema ⚠
transaction-events	--	3	--	--	1.59KB	--	Add schema ⚠

transaction-events (3 partitions) and notification-events (1 partition)

2.2. Publish Approved Event



HDFC MF / Kafka POC / Transaction Event Approved

POST http://localhost:8090/api/events/approved

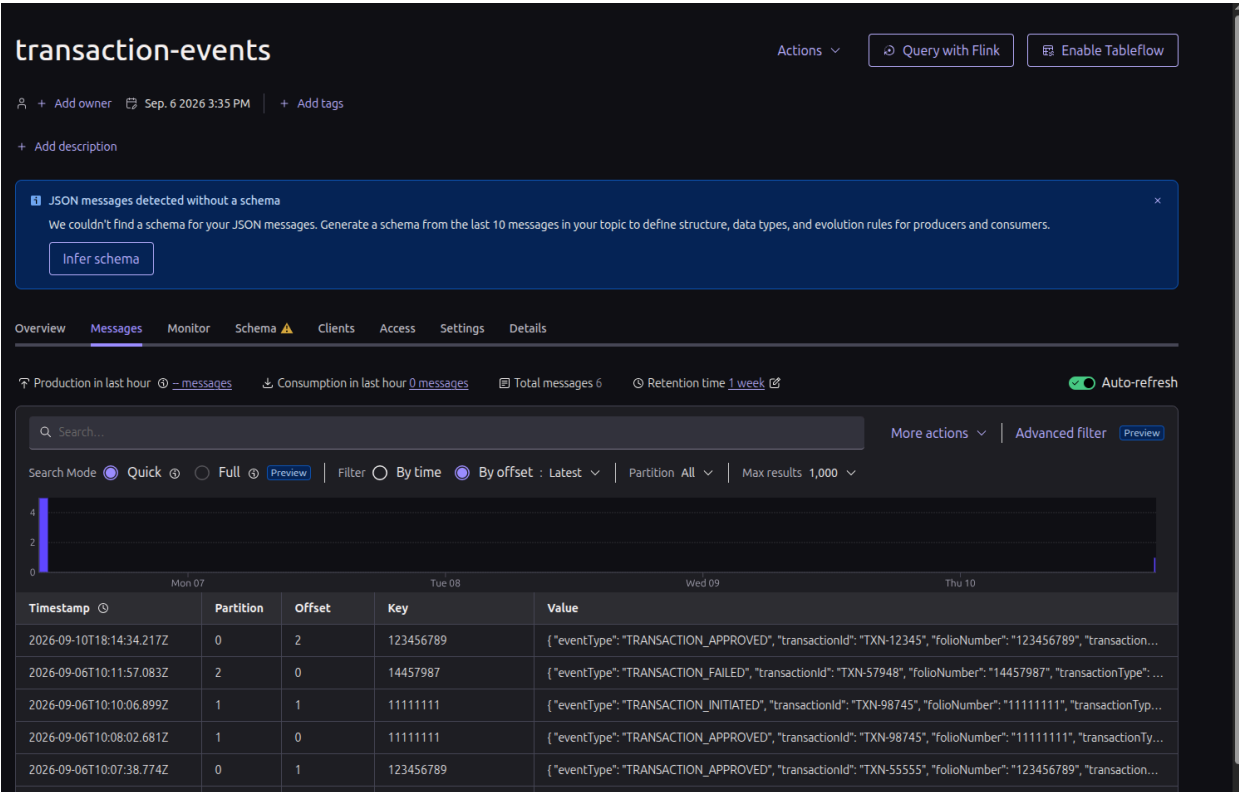
Body

```
1 {
2   "transactionId": "TXN-12345",
3   "folioNumber": "123456789",
4   "transactionType": "REDEMPTION",
5   "phoneNo": "9999999999"
6 }
```

202 Accepted · 788 ms · 218 B

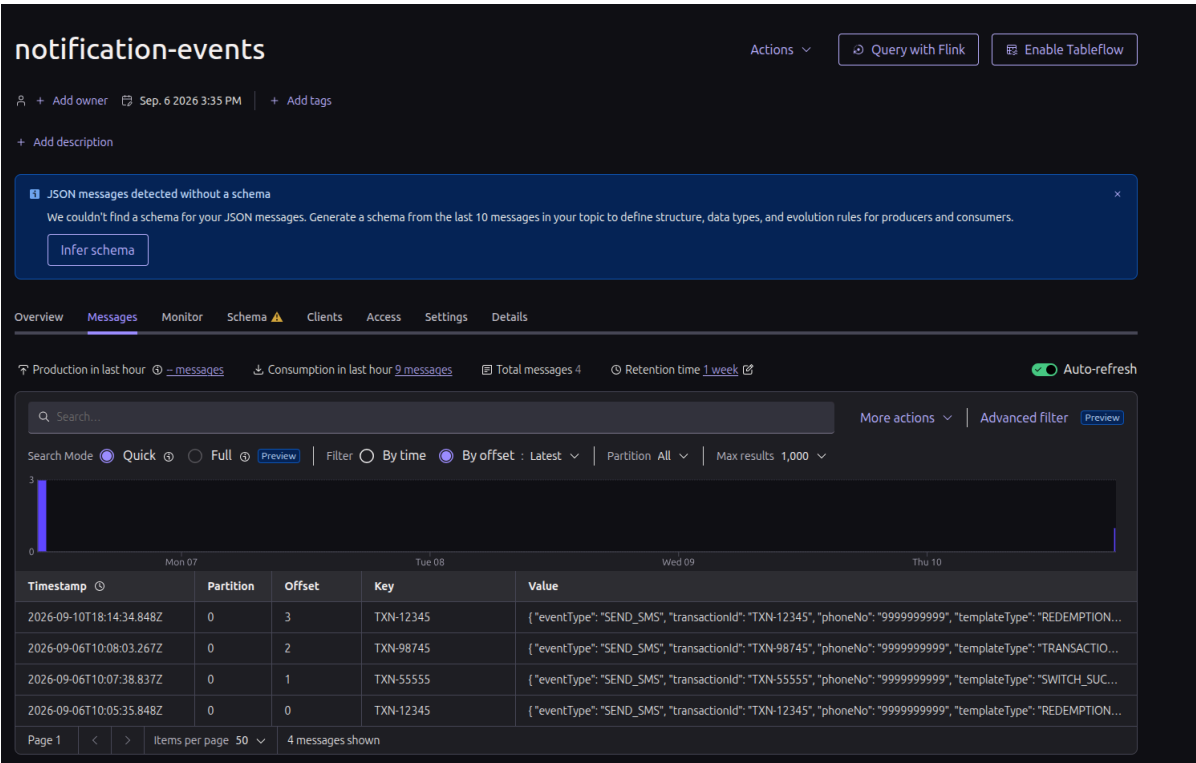
1 Published TRANSACTION_APPROVED for txn=TXN-12345

2.3. transaction-events Messages



APPROVED, INITIATED, FAILED events in Confluent Cloud

2.4. notification-events Messages



SEND_SMS events published after approval

2.5. Kafka Configuration

```
application.yml
1  server:
2    port: 8090
3
4  spring:
5    config:
6      import:
7        - classpath:env.properties
8    application:
9      name: kafka-poc
10  kafka:
11    bootstrap-servers: ${KAFKA_BOOTSTRAP_SERVERS}
12    security:
13      protocol: SASL_SSL
14    properties:
15      sasl:
16        mechanism: PLAIN
17      jaas:
18        config: org.apache.kafka.common.security.plain.PlainLoginModule required username="${KAFKA_API_KEY}" password="${KAFKA_API_SECRET}";
19    producer:
20      key-serializer: org.apache.kafka.common.serialization.StringSerializer
21      value-serializer: org.springframework.kafka.support.serializer.JsonSerializer
22    consumer:
23      group-id: kafka-poc-default
24      key-deserializer: org.apache.kafka.common.serialization.StringDeserializer
25      value-deserializer: org.springframework.kafka.support.serializer.ErrorHandlingDeserializer
26      properties:
27        spring.deserializer.value.delegate.class: org.springframework.kafka.support.serializer.JsonDeserializer
28        spring.json.trusted.packages: com.learningkafka.kafkapoc.model
29
30  logging:
31    level:
32      com.learningkafka.kafkapoc: INFO
```

2.6. Consumer Logs - Approved Flow

```
INFO 624400 --- [kafka-poc] [nio-8090-exec-2] o.a.k.c.t.i.KafkaMetricsCollector : Initializing Kafka metrics collector
INFO 624400 --- [kafka-poc] [nio-8090-exec-2] o.a.k.c.producer.KafkaProducer : [Producer clientId=kafka-poc-producer-1] Instantiated an idempotent producer.
INFO 624400 --- [kafka-poc] [nio-8090-exec-2] o.a.kafka.common.utils.AppInfoParser : Kafka version: 3.7.1
INFO 624400 --- [kafka-poc] [nio-8090-exec-2] o.a.kafka.common.utils.AppInfoParser : Kafka commitId: e2494e6ff89f828
INFO 624400 --- [kafka-poc] [nio-8090-exec-2] o.a.kafka.common.utils.AppInfoParser : Kafka startTimeMs: 1789064073899
INFO 624400 --- [kafka-poc] [-poc-producer-1] org.apache.kafka.clients.Metadata : [Producer clientId=kafka-poc-producer-1] Cluster ID: lkc-81gmkp0
INFO 624400 --- [kafka-poc] [-poc-producer-1] o.a.k.c.p.internals.TransactionManager : [Producer clientId=kafka-poc-producer-1] ProducerId set to 28002 with epoch 0
INFO 624400 --- [kafka-poc] [-poc-producer-1] o.a.k.c.t.i.ClientTelemetryReporter : Client telemetry registered with client instance id: oSllTDgWTem8FQxeCqSP2Q
INFO 624400 --- [kafka-poc] [ntainer#0-0-C-1] c.l.k.consumer.AuditEventConsumer : AUDIT: TRANSACTION_APPROVED txn=TXN-12345 folio=123456789 type=REDEMPTION
INFO 624400 --- [kafka-poc] [ntainer#0-0-C-1] c.l.k.p.NotificationEventPublisher : Publishing SEND_SMS for txn=TXN-12345 to notification-events
INFO 624400 --- [kafka-poc] [ntainer#1-0-C-1] c.l.k.c.NotificationEventConsumer : SMS sent (simulated) to 9999999999 template=REDEMPTION_SUCCESS params={transactionType=REDEMPTION, folioNumber=123456789}
```

SEND_SMS events published after approval

2.7. Consumer Logs - Initiated Event

```
2026-09-10T23:44:33.849+05:30 INFO 624400 --- [kafka-poc] [nio-8090-exec-2] o.a.k.c.t.i.KafkaMetricsCollector : initializing Kafka metrics collector
2026-09-10T23:44:33.865+05:30 INFO 624400 --- [kafka-poc] [nio-8090-exec-2] o.a.k.clients.producer.KafkaProducer : [Producer clientId=kafka-poc-producer-1] Instantiated an idempotent producer.
2026-09-10T23:44:33.899+05:30 INFO 624400 --- [kafka-poc] [nio-8090-exec-2] o.a.kafka.common.utils.AppInfoParser : Kafka version: 3.7.1
2026-09-10T23:44:33.900+05:30 INFO 624400 --- [kafka-poc] [nio-8090-exec-2] o.a.kafka.common.utils.AppInfoParser : Kafka commitId: e2494e6ffb89f828
2026-09-10T23:44:33.900+05:30 INFO 624400 --- [kafka-poc] [nio-8090-exec-2] o.a.kafka.common.utils.AppInfoParser : Kafka startTimeMs: 1789864873899
2026-09-10T23:44:34.217+05:30 INFO 624400 --- [kafka-poc] [-poc-producer-1] org.apache.kafka.clients.Metadata : [Producer clientId=kafka-poc-producer-1] Cluster ID: lkc-8lgnkp0
2026-09-10T23:44:34.218+05:30 INFO 624400 --- [kafka-poc] [-poc-producer-1] o.a.k.c.p.internals.TransactionManager : [Producer clientId=kafka-poc-producer-1] ProducerId set to 28002 with epoch 0
2026-09-10T23:44:34.219+05:30 INFO 624400 --- [kafka-poc] [-poc-producer-1] o.a.k.c.t.i.ClientTelemetryReporter : Client telemetry registered with client instance id: oSllTDgMTem8FQxeCq5PZQ
2026-09-10T23:44:34.815+05:30 INFO 624400 --- [kafka-poc] [ntainer#0-0-C-1] c.l.k.consumer.AuditEventConsumer : AUDIT: TRANSACTION_APPROVED txn=TXN-12345 folio=123456789 type=REDEMPTION
2026-09-10T23:44:34.816+05:30 INFO 624400 --- [kafka-poc] [ntainer#0-0-C-1] c.l.k.p.NotificationEventPublisher : Publishing SEND_SMS for txn=TXN-12345 to notification-events
2026-09-10T23:44:35.121+05:30 INFO 624400 --- [kafka-poc] [ntainer#1-0-C-1] c.l.k.c.NotificationEventConsumer : SMS sent (simulated) to 9999999999 template=REDEMPTION_SUCCESS params={transactionType=REDE
2026-09-10T23:48:10.471+05:30 INFO 624400 --- [kafka-poc] [nio-8090-exec-5] c.l.k.p.TransactionEventPublisher : Publishing TRANSACTION_INITIATED for txn=TXN-98745 folio=11111111 to transaction-events
2026-09-10T23:48:11.370+05:30 INFO 624400 --- [kafka-poc] [ntainer#0-0-C-1] c.l.k.consumer.AuditEventConsumer : AUDIT: TRANSACTION_INITIATED txn=TXN-98745 folio=11111111 type=SIP
```

INITIATED event audited (no notification)

3. GitHub Repository

Repository: <https://github.com/ananya-shukla-TTN/kafka-poc>

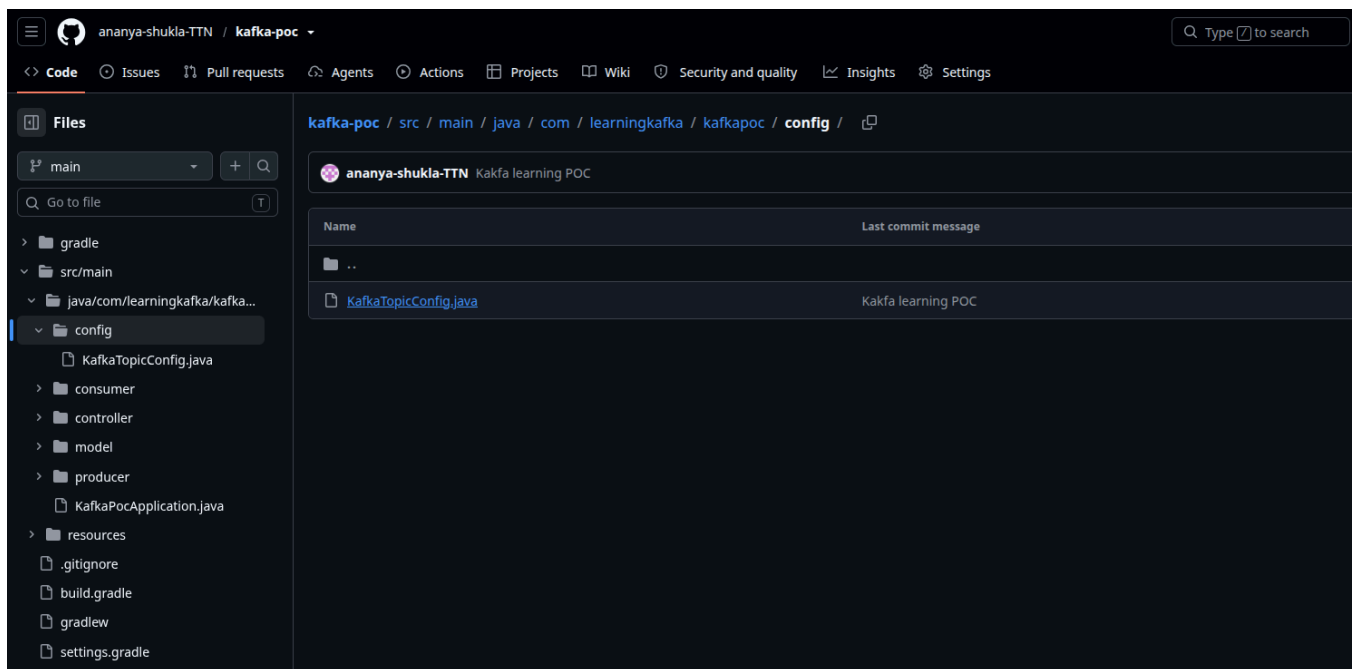


Figure: GitHub Repository Screenshot